

Seat No: _____

No. of printed pages: 2

[80]

SARDAR PATEL UNIVERSITY

B.C.A. (SEMESTER- 3) (On Demand), February 2026

US03CBCA51: Object Oriented Programming - I

Date: 10-02-2026, Tuesday

Time: 2.00 pm – 5.00 pm



Marks: 70

Q-1 Select the correct option from the following questions. (10)

- 1) A _____ is a collection of objects of similar type.
(a) Data (b) Class (c) Object (d) Function
- 2) _____ is the developer of C++.
(a) Dennis Ritchie (b) Ivan Byros (c) Bjarne Strastrup (d) James Gosling
- 3) _____ is known as scope resolution operator.
(a) << (b) >> (c) & (d) ::
- 4) _____ is a selection control structure.
(a) if (b) switch (c) if ... else (d) All of Above
- 5) _____ function is used to copy one string into another string.
(a) strcpy() (b) stringcpy() (c) strcopy() (d) StrCopy()
- 6) While defining the derived class, base class name is written on _____ side of : symbol.
(a) LHS (b) RHS (c) Both (d) Any one
- 7) In case of multipath inheritance, problem of duplicate data-members is resolved using _____ keyword.
(a) abstract (b) const (c) virtual (d) default
- 8) Which one of the following is not a valid argument passing mechanism in C++?
(a) pass by value (b) pass by selection (c) pass by reference (d) pass by address
- 9) Which one of the following operators can be overloaded?
(a) New (b) Sizeof (c) :: (d) .*
- 10) The correct function name for overloading the addition (+) operator is
(a) operator(+) (b) operator+ (c) +operator (d) operator:+

Q-2 Short Questions (Attempt any Ten) (20)

- 1) List out any four header files in C++.
- 2) Define encapsulation as concept of OOP.
- 3) Define class & object.
- 4) List all access specifiers and also write its use.
- 5) Define destructor and give its example.
- 6) Define string and write how to declare string in C++.

[1]

(P.T.O)

- 7) What is the difference between private & protected members of the class?
- 8) Define inheritance & state two advantages of it.
- 9) What is a virtual base-class? What is use of it?
- 10) What is a stream?
- 11) What is an operator function? Describe syntax of an operator function.
- 12) List some of the rules for overloading operator.

Q-3 A Explain different operators available in C++. (05)

B What is C++? Explain structure of C++. (05)

OR

Q-3 A Explain different data type available in C++. (05)

B Differentiate OOP & POP. (05)

Q-4 A Explain basic I/O in C++ with proper example. (05)

B What do you mean by constructors? What is the use of constructors in C++? (05)

OR

Q-4 A Explain constant and nameless object with example. (05)

B Explain declaration and initialization of one dimensional array with example. (05)

Q-5 A Explain multiple and multilevel inheritance with example. (05)

B Describe the concept of friend function with suitable example. (05)

OR

Q-5 A Discuss virtual function with example. (05)

B Discuss inline functions with proper example. (05)

Q-6 A Explain Unary operator overloading with example. (05)

B What is file mode? Describe various file mode operations available. (05)

OR

Q-6 A Explain Binary operator overloading with example. (05)

B Explain opening file using open(). (05)